Developing minds in a digital world, a changing India



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Each evening, a familiar scene unfolds in homes around the world. A teenager moves seamlessly between chats, videos, social media, and games on their phone. Homework is completed using the latest Artificial Intelligence (AI) platform. Screens increasingly dominate everyday life. This is a profound change in how developing minds experience the world, learn, and grow.

These changes are particularly significant for India, with approximately 65 crore people under the age of 25 and one of the world's largest and fastest-growing technological ecosystems. Digital engagement by this vast young population continues to grow as education, recreation, and social interaction increasingly move online.

The digital world has brought undeniable benefits, from unprecedented access to information and educational resources to new ways of communicating, creating, and learning. Recent advances in AI have expanded these possibilities even further. Nonetheless, these digital environments have also introduced a powerful and largely unregulated force into children's lives – one we are only beginning to understand.

A brain under construction

Technology itself is not the problem. The concern is constant exposure to powerful digital tools during critical periods of development, when the brain systems underlying cognition, emotion, and behaviour are still maturing. These concerns extend beyond childhood. The prefrontal cortex – which supports attention, working memory, planning, decision-making, and impulse control – continues to mature into early adulthood.

Digital environments provide a dense, repetitive stream of experiences that can shape how brain circuits develop – strengthening some pathways while leaving others underdeveloped. Given the brain's plasticity during early life, such patterns may have lasting effects, shaping how young people think, feel, and act across the lifespan.

Emerging research links high levels of screen time to alterations in brain structure and connectivity. Though revealing, such findings do not establish causation. They do, however, underscore the need to better understand how digital experiences affect cognitive development, including attention, memory, self-control, and creativity.

Social media platforms are particularly appealing during adolescence, when belonging, peer evaluation, and social approval are especially salient. These platforms are designed to capture and sustain attention by exploiting basic features of cognition, emotion, and reward. Short videos, autoplay, infinite scroll, algorithmic recommendations, and social feedback create reinforcement loops that deliver novelty and rewards – likes, shares, and comments – in a rapid but unpredictable manner.

This form of variable reinforcement encourages repetitive checking and rapid attention shifts, potentially making it harder to sustain attention on a single task. Over time, such patterns of engagement may shape what young people come to expect from everyday experiences. When digital rewards are immediate and constantly accessible, ordinary experiences – having a face-to-face conversation, practising a skill, or simply sitting with one's thoughts – may feel less rewarding by comparison.

Offloading mental effort

Generative AI has rapidly transformed how we engage with the world. Unlike earlier technologies, AI does not just assist with mental effort; it can supplant it. Humans have long used tools to reduce cognitive load. Taking a photograph in a crowded parking lot, for example, means we do not have to remember where we parked. Such cognitive offloading is not inherently problematic.

The concern arises when AI replaces the mental effort necessary for learning. Students can now use AI platforms to generate ideas, organise arguments, write essays, and summarise material. Used thoughtfully, these tools can enhance the learning process. When used as a substitute for independent thinking, however, AI can undermine the cognitive effort through which knowledge and skills develop.

Grappling with difficult questions, finding the right words, making mistakes, and revising arguments are central to learning. If AI routinely eliminates these experiences, we should rightly ask what the impact will be on young people's ability to think critically, use informed judgement, and persevere when answers are not immediately available.

A related concern is what happens when technology replaces not just our intellectual efforts but our idle moments. Though rarely welcomed, boredom creates space for imagination, reflection, creativity, and the development of a rich inner life. When every free moment is filled with stimulation, young people have fewer opportunities to think, wonder, daydream, or reflect.

Neuroplasticity – the brain's lifelong capacity to adapt to experience, learning, and the environment – offers another perspective on how digital technology shapes young minds. Today's youth are certainly adapting to a world very different from that of previous generations. Indeed, many young people exhibit impressive abilities to filter information, navigate complex digital environments, and switch among information sources.

We must be cautious, however, when evaluating whether these changes are beneficial or harmful. It is important to consider what may be gained – and lost – as young minds adapt to today's technologically advanced ecosystem.

In effect, young people worldwide are participating in a vast, largely uncontrolled experiment in how sustained exposure to digital technology may shape developing brains and minds. Meanwhile, the corporations designing these digital tools and devices have strong financial incentives to keep young people – their current and future users – engaged.

Safeguarding the future

India's youth are growing up in an environment that was unimaginable until quite recently. Whatever our concerns, we cannot simply ignore or reject digital technology. Children will use it. Our responsibility is to help them use it capably, thoughtfully, and responsibly.

Parents need practical, evidence-based guidance. Children should have device-free times and places, protected study periods, healthy sleep routines, and opportunities for unstructured time. The objective is not to eliminate screens, but to ensure that they do not occupy every available moment.

Schools and educators must preserve what might be called productive struggle – the effort involved in thinking, writing, reasoning, solving problems, and working through uncertainty. Digital tools, including AI, can support analysis, revision, and feedback without replacing critical thinking. Greater emphasis should be placed on independent cognitive engagement, including oral examinations, supervised writing, and applied problem-solving.

Digital platforms – especially those targeted at young people – must face greater scrutiny and accountability. Policymakers should require companies to provide stronger evidence about how their products affect the cognitive and emotional development of children and adolescents. Governments must also support independent research capable of informing policies and regulations as these technologies continue to evolve.

We should not wait for definitive evidence of harm before asking whether the digital environments we are creating are conducive to healthy development. A young mind needs stimulation, but also silence; information, but also reflection; assistance, but also struggle. We can embrace technological progress while still protecting the experiences through which children and adolescents learn to think, imagine, reflect, and grow.

Digital innovation must be balanced with children's cognitive, emotional and social development

A case for joining the global carbon coalition



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On July 31, nearly 500 Indian companies across seven industrial sectors filed their first-ever compliance reports under the national Carbon Credit Trading Scheme (CCTS). The country's scheme – on paper since 2023 – is now live with binding targets, penalties and real entities. The first credits are due to be issued in October, with trading on power exchanges expected to follow between November and January. In May, the European Union, China, and Brazil launched the Open Coalition on Compliance Carbon Markets (OCCCM) in Florence, aimed at aligning the way the world's compliance carbon markets communicate with each other. Germany and New Zealand joined as the first new members; Canada, France, Norway, Singapore, Türkiye and the United Kingdom have since followed. On September 14, at its meeting in Wuhan, the coalition adopted its first work plan.

India has stayed outside so far. With its own market now live, the more interesting question is no longer whether India can afford to build a carbon price – it already has. It is whether India can afford to sit outside the room where its rules get written. Or is it best avoided due to unforeseen risks? The OCCCM is a collaborative initiative for exchange on carbon pricing mechanisms; monitoring, reporting and verification (MRV) systems, carbon accounting methodologies, and rules related to the potential use of high-integrity offsets.

In India, the government has adopted a compliance carbon market framework in the form of a Carbon Credit Trading Scheme (CCTS), which is currently in its first phase. The CCTS (compliance) is set up to deliver on India's Nationally Determined Contribution (NDC) targets and the 2070 net-zero goal.

The issue of a carbon price

There are two key reasons why India must seriously consider joining the OCCCM. The first is the proliferation of carbon pricing systems worldwide. The idea of having a carbon price as a key instrument, supported by complementary policy instruments, is back at the high table. It minimises the economy-wide cost of mitigation (critical for the global South) and can promote innovation while minimising the impact on the vulnerable. It can also deliver significant fiscal revenues at a time when government resources are increasingly stretched, including by the impacts of climate change – from heatwaves to devastating floods. As of now, over 50 countries have instituted carbon pricing measures. As the world advances towards a net-zero future, the likelihood that more and more countries adopt a carbon price is high.

We could be looking at a future where a bulk of emissions would be priced. At some point, more and more of these domestic carbon pricing systems will be linked – even if partially – to better coordinate. While the beginning of OCCCM focuses on knowledge exchange, one potential long-term strategic outcome is improved interoperability among domestic carbon pricing systems. It could be valuable to link India's CCTS (compliance) to other carbon trading systems, given the potential to earn revenue from jurisdictions with more stringent mitigation targets.

On 'climate clubs'

The second is to be better prepared in case a coalition along the lines of the EU's Carbon Border Adjustment Mechanism (CBAM) develops. The idea of 'climate clubs' is re-emerging as geopolitics fractures United Nations Framework Convention on Climate Change (UNFCCC) discussions and attention turns to implementation. A report on a climate coalition centred around the CBAM issue was released in the context of an independent council convened by the COP30 President, Ambassador André Aranha Corrêa do Lago (The writers of this article are the co-authors of the report). The report describes alternative scenarios for CBAM coalitions in which there could be uniform or differentiated carbon prices across member-countries, incentives for member-countries, and penalties for non-member countries. With the European Union CBAM now operational, and many other countries contemplating their own versions of CBAM, being better prepared on this front is absolutely critical for India. While the OCCCM does not mention CBAM explicitly – most probably because of the sharp reactions the term evokes, especially during climate negotiations – OCCCM's objectives around the MRV, carbon accounting, and domestic compliance markets are very relevant for being better prepared to deal with the CBAM world.

Why joining matters now

Are there potential risks in joining the coalition? The coalition is about better alignment of domestic compliance markets and carbon pricing across countries. It does not seek to enforce any mitigation targets or any other decision. However, as discussions evolve, some critical decisions are likely to be made by coalition members. OCCCM's governance structure consists of high-level meetings (the highest decision-making body), executive-level meetings, thematic working groups and a Secretariat. It is critical to be in the room and shape the discussions rather than respond to the developments as an outsider. The risk is more in being an outsider. That Brazil is a founding member, and other like-minded developing countries are in the process of becoming members, should be a source of comfort for India.

Some aspects of the future world are likely: geopolitics will remain fractured, climate negotiations will continue to meet logjams, the profile of carbon pricing will continue to rise across countries, and efforts from outside the Paris Agreement framework will become increasingly important to achieving the agreement's goals. In this context, the role played by country-backed climate coalitions will likely grow.

The OCCCM aligns well with India's decarbonisation choices and offers a useful platform not just to learn about carbon markets, but also to be better prepared to engage with other national carbon pricing architectures and border carbon taxes within the evolving geopolitical landscape. India should consider making this move sooner rather than later.

The views expressed are personal

India should join the Open Coalition on Compliance Carbon Markets, as it aligns well with the country's decarbonisation choices

Revamp functioning of Parliament



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India's impending delimitation debate may appear, at first glance, to be a matter of democratic arithmetic: if too many citizens are being represented by one Member of Parliament (MP), then the answer seems to be to simply increase the number of MPs. However, in a diverse democracy like India, representation cannot be limited to the number of MPs occupying the House; rather, it must also ensure that those elected have the institutional space to speak for their constituents.

Parliamentary representation operates in two layers: the voter's access to their MP and the MP's ability to speak for them on the floor of the House. The latter relies significantly on whether the MP receives the actual floor time to raise issues. While delimitation may bring legislators closer to their constituents, merely increasing the seats without strengthening their functional capacity will leave representatives muted on the floor.

The jointly introduced Constitution (131st Amendment) Bill and the Delimitation Bill, 2026, provide for increasing the strength of the Lok Sabha from 550 to 850 members. As per the Election Commission's 2024 Atlas, India manages an electoral roll of 97.98 crore voters, across 543 Lok Sabha constituencies, averaging roughly 18 lakh voters/constituency, highlighting the burden on an MP. This average also conceals severe constituency-wide disparities: the constituency of Malkajgiri in Telangana has 37.8 lakh voters, whereas Lakshadweep has just 57,953. While these realities signify an urgent need for restructuring, the proposed plan has turned a blind eye towards increasing Parliament's working capacity.

This missing institutional capacity represents a major structural blind spot. Adding hundreds of members will worsen an existing problem: the severe lack of available floor time. If Parliament's operational timetable remains fixed, adding more lawmakers dilutes individual members' ability to participate in legislative business.

If Parliament's operational timetable remains fixed, adding more lawmakers dilutes individual members' ability to participate in legislative business

Not enough time

In the Question hour currently, Lok Sabha rules allow a maximum of 20 starred questions for oral answers and 230 unstarred questions for written answers per sitting day. Even though each MP can submit up to five questions daily, only a tiny fraction is selected through the daily ballot. If the total number of MPs is increased to 850, while keeping daily sitting hours the same, the chances of an MP getting an oral answer from a Minister will become even more low.

Moreover, during the 17th Lok Sabha, Question Hour functioned for only 60% of its scheduled time. The time constraints were even more severe during the 2026 Monsoon Session, where the Question Hour operated for only 1% of its scheduled time, running for a total of just nine minutes across the entire session. Adding more members to a House operating under these constraints will not improve executive-legislative accountability.

Other crucial parliamentary procedures will also be impacted by this reduction in time. For example, MPs use Zero Hour to raise urgent issues as it does not require any advance notice. It has been crucial in bringing to light urgent public matters such as regional crop failures or disaster relief. Similarly, Private Member's Business gives MPs who are not part of the Cabinet a chance to bring in their own legislation. These are indispensable instruments that will lose their essence if they are not protected with measures that allow them to be fully utilised.

In order to protect the voice of the representatives, it is necessary to give Parliament's sitting time a statutory status. Article 85(1) of the Constitution states that the President shall summon each House of Parliament to meet at such time and place as he thinks fit, but six months shall not intervene between its last sitting in one session and the date appointed for its first sitting in the next session. This provision should be expanded to include a mandatory minimum working days clause. As recommended by the National Commission to Review the Working of the Constitution, Lok Sabha must meet for a minimum of 120 days per year. Additionally, parliamentary instruments such as Zero Hour and Question Hour need protected status, with a mandatory allocated time that cannot be changed as per the directions of the Speaker.

While delimitation can redraw the boundaries of the constituencies, parliamentary reforms are necessary to ensure that the voices within those constituencies are heard.

Analysing female labour of the North East States and UTs

While the geography of female labour force participation is changing, it is not at a uniform pace

DATA POINT

Chandrasekar K.

The Periodic Labour Force Survey (PLFS) 2025 report shows a clear increase in Female Labour Force Participation (FLFPR) since 2019-20. The rise was more pronounced in rural areas, where the percentage increased from 33% to 45.9%, than in urban areas, where it rose from 23.3% to 27.7%. This difference calls for a closer look at the northeastern States, where female labour force participation is higher, and Union Territories (UTs) to understand how female labour force participation changed in rural and urban areas.

Chart 1 presents the Average Annual Percentage Point (AAPP) change in rural FLFPR for each State between 2019-20 and 2025. It shows that major rural northeastern States and UTs are above this benchmark. Arunachal Pradesh records the highest increase, at 8.28 percentage points per year, more than three times the all-India benchmark of 2.58. At the other end, FLFPR declined in Delhi and Lakshadweep. The difference between the highest and lowest AAPP is more than 10 percentage points, from +8.28 in Arunachal Pradesh to −2.34 in Lakshadweep. This means that the national average of 2.58 points conceals substantial State-level variation.

Chart 2 shows that Arunachal Pradesh stands out again, with an AAPP of 5.24 points above the all-India urban AAPP of 0.88 pp, followed by Meghalaya, Nagaland, Manipur, Chandigarh, Andaman & Nicobar Islands, Assam, Tripura and J&K. But Puducherry, Ladakh, Delhi and Lakshadweep recorded negative AAPP. The urban range, from +5.24 to −1.30 percentage points, remains considerable, even though the national average is much lower than in rural areas.

The rural all-India AAPP is therefore 1.7 percentage points higher per year than the urban figure. For a statistical analysis, this is important: the national direction and the State-level distribution are two different stories. While the aggregate shows an increase, it does so across the NE States, while the UTs are far more heterogeneous.

However, AAPP alone is insufficient to evaluate the performance of northeastern States and UTs, as it measures only the pace of change and does not consider the level of female labour force participation from which a State started. Therefore, the 2019-20 FLFPR for each State is used as the baseline. Combining the baseline level with the subsequent AAPP provides a more meaningful assessment of whether States are catching up from a lower base, sustaining improvements at higher levels, or making slow progress.

In **Chart 3**, the upper-left portion of the rural scatter contains Arunachal Pradesh, Assam, Manipur and Tripura. These States had an FLFPR below the all-India 2019-20 benchmark but recorded AAPP above the national average, indicating faster improvement, whereas Delhi recorded an AAPP below the all-India average. At the other end, Nagaland, Meghalaya, Andaman & Nicobar and J&K began with their FLFPR above benchmark and recorded AAPP above the national average. This indicates that faster improvement is not confined to States starting from a low base.

Chart 4 shows that in urban areas, Arunachal Pradesh, Chandigarh, Assam and Tripura started from relatively low baseline levels but recorded AAPP above the all-India average, indicating relatively faster improvement. In contrast, Delhi also started from relatively low levels but recorded below-average AAPP. Among States with relatively higher baselines, Meghalaya, Nagaland, Manipur, and J&K had above-average AAPP, showing that some States with already higher participation levels continued to make relatively rapid gains.

Measuring the change

The data for the charts were sourced from PLFS 2019 and 2025

The writer is an Indian Statistical Service Officer. Views expressed are personal



CHART 1: The Average Annual Percentage Point change (AAPP) in rural FLFPR for each State between 2019–20 and 2025

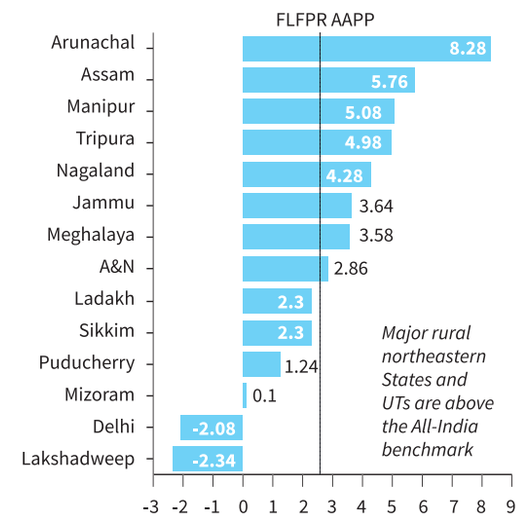


CHART 2: The Average Annual Percentage Point change (AAPP) in urban FLFPR for each State between 2019–20 and 2025

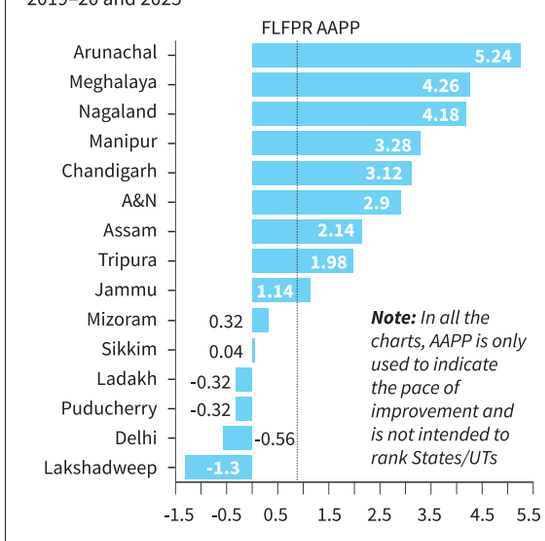


CHART 3: Comparing subsequent AAPP against the 2019-20 LFPR benchmark for rural areas. Cut-off lines represent the corresponding All-India 2019-20 LFPR and AAPP

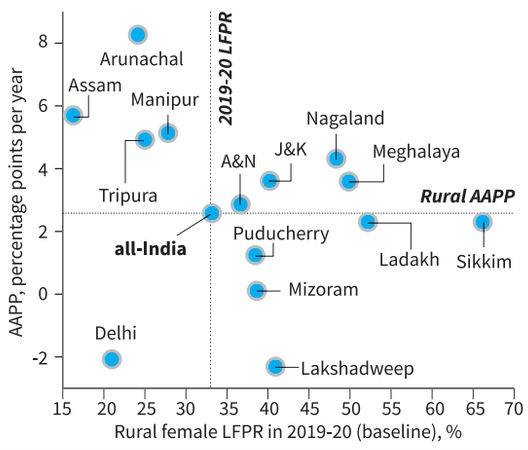
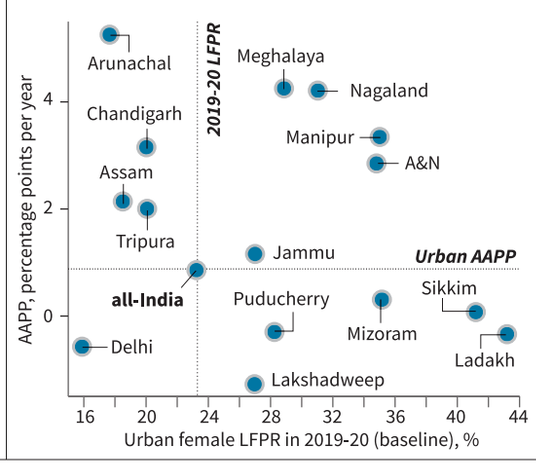


CHART 4: Comparing subsequent AAPP against the 2019-20 LFPR benchmark for urban areas. Cut-off lines represent the corresponding All-India 2019-20 LFPR and AAPP



Text&Context

THE HINDU

Separation of powers and CJI's refusal

Why did the Opposition seek a meeting with the CJI? What allegations did it raise against the CEC? Why did the CJI refuse the request? What remedies did Rahul Gandhi seek from the Supreme Court? How does separation of powers apply to the case? What does judicial propriety require of judges?

EXPLAINER

Krishnadas Rajagopal

The story so far:

Chief Justice of India Surya Kant declined a request made by Opposition leaders for an audience to discuss allegations against Chief Election Commissioner (CEC) Gyanesh Kumar's conduct during the Special Intensive Revision (SIR) process, alongside detentions of peaceful protestors and the lockdown of arterial Delhi.

The refusal was anchored in the ethos of maintaining a neutral distance from the political thicket. The CJI's letter emphasised that the judiciary cannot blur constitutional lines.

"A short while ago, the One-Nation-One-Election Joint Parliamentary Committee (JPC) had proposed to meet the judges of the Supreme Court. The leaders of the Bar had questioned how parliamentarians could meet judges when the Bill on simultaneous elections was itself under consideration. The law, which may emanate from the Bill, could one day, down the line, come up before the court in challenge. So, in principle, I don't see how the Chief Justice was wrong in applying the same standard about meeting Mr. Rahul Gandhi as was applied in the instance of the JPC. It is a matter of propriety," senior advocate C.U. Singh said.



GETTY IMAGES

THE GIST

- The CJI refused to meet Opposition leaders Rahul Gandhi and Mallikarjun Kharge over allegations against the CEC during the SIR, citing separation of powers and judicial propriety
- He advised them to seek judicial remedies, noting that the issues were already before the court and that the Supreme Court could hear petitions to enforce citizens' fundamental rights
- Past controversies involving judges and political leaders show concerns over judicial neutrality and the need to insulate the judiciary from executive influence

Why did the Opposition approach the CJI?

On October 10, the Leaders of the Opposition in the Rajya Sabha and Lok Sabha, Mallikarjun Kharge and Rahul Gandhi, respectively, wrote to CJI Kant. It marked a rare and historic instance of the Opposition directly engaging the highest judicial office of the country to complain against another constitutional authority, the CEC.

The brief request, submitted on a joint letterhead, stated that the two leaders wished to submit a memorandum to the Chief Justice concerning the CEC and the Election Commission of India. Sent hours after the brutal police crackdown on citizens demanding the resignation of CEC Gyanesh Kumar, the representation said the arbitrary arrest of students, closure of Metro stations, cancellation of trains, and barricading of arterial roads created an "atmosphere of siege in the city of Delhi".

Mr. Gandhi followed up with a second, detailed representation to CJI Kant on October 11. He underscored the Supreme Court's role as the guardian of the Constitution and the fundamental right of every citizen to vote. Writing in his capacity as the Leader of the Opposition in the Lok Sabha on behalf of crores of voters anxious about the integrity of the electoral rolls, Mr. Gandhi pointed out that he was also a member of the Selection Committee constituted under Section 7 of the Chief Election Commissioner and Other Election Commissioners (Appointment, Conditions of Service and Term of Office) Act, 2023. He underscored that he had recorded a formal note of dissent on the procedure followed for selecting CEC Kumar during a committee meeting on February 17, 2025.

The representation accused the CEC of an arbitrary and unilateral exercise of power during the SIR. He asked the CJI to treat the representation as a letter petition in the public interest.

He urged the CJI to direct a complete disclosure of SIR records with the Election Commission and the setting of an independent committee headed by a retired Supreme Court judge to review decisions and audit voter deletions made during SIR to safeguard free and fair elections. He sought an interim suspension of the SIR process itself.

How did the CJI respond?

On October 12, the Chief Justice responded to the October 10 request for a meeting, while omitting any direct reference to Mr. Gandhi's subsequent October 11 representation.

Addressing both Mr. Kharge and Mr. Gandhi and declining their request for an appointment on October 12 between 4 pm and 5 pm to submit a memorandum, the CJI noted that the issues regarding the CEC, the arbitrary arrests, Metro closure and train cancellations, etc., were already sub judice before the court.

The CJI said that, "therefore, convening a meeting to discuss pending cases would wholly contravene the principle of Separation of Powers, an intrinsic facet of the basic structure doctrine of the Constitution of India, and would be incompatible with the established ethics and judicial propriety".

The two-page response also reminded the leaders that the constitutional framework did not vest the Supreme Court with administrative powers to "deal with the issues highlighted in your letter". But the CJI said the Supreme Court was duty-bound on the judicial side to consider the gravamen of issues affecting the citizens of India.

The top judge reiterated in his reply to the Opposition leaders the court's commitment to preserve and protect the fundamental rights of every citizen of India.

Chief Justice Kant said citizens were "entitled to seek enforcement of such rights at all hours through the procedure known in law".

Chief Justice Kant advised the two leaders in his response that it would be "more appropriate on your part to invoke the judicial powers of the Supreme Court of India, instead of submitting a memorandum on the administrative side".

The CJI said that petitions raising parallel issues were listed for hearing on October 13. He said the leader, acting on legal advice, could file their petition on October 12. The CJI observed that the leaders could file their petition on October 12 based on legal advice, assuring that "any such petition shall be heard at the earliest possible opportunity, in accordance with law". A writ petition was subsequently filed by Mr. Gandhi alongside 12 other Opposition leaders, which was successfully registered, numbered, and is primed for a judicial hearing.

What does the law say?

At the heart of the CJI's response lie three factors – the doctrine of separation of powers, sub judice rules and judicial propriety, and the Bangalore Principles.

As Chief Justice Kant's response said, the separation of powers, which divides governance among the three distinct organs of the legislature, the executive, and the judiciary, was an unalterable part of the basic structure of the Constitution. The court functions as an impartial arbiter with a clear mandate to adjudicate disputes impartially, apply laws, and review executive or legislative actions only through established judicial proceedings. The judiciary did not hold executive or administrative oversight to unilaterally intervene in government policy or governance demands.

A parallel out-of-court meeting between the Chief Justice, who also happens to lead the Bench hearing the CEC-related cases and Opposition leaders, belies fair play and bypasses procedural fairness. Entertaining political factions on a one-to-one basis outside the courtroom compromises judicial impartiality and creates an impression of bias. Disputes must be settled in the open court, transparently, in proceedings in which opposing parties get an equal say in accordance with the principles of natural justice.

The Bangalore Principles of Judicial Conduct puts in place ethical benchmarks for judges, which include that they must avoid any act of impropriety which would compromise the public's perception of judicial neutrality.

When has judicial propriety come under scrutiny?

Prime Minister Narendra Modi's 2024 visit to the residence of then Chief Justice of India, Justice Dhananjay Chandrachud, to perform pooja during Ganesh Chaturthi drew public criticism over perceptions of the executive and judiciary cosying up. Historical precedents of serving judges expressing admiration for political figures include Justice P.N. Bhagwati sending a congratulatory letter to Indira Gandhi upon her return to power, and Justice M.R. Shah reportedly describing Prime Minister Modi as a "model and a hero". Similarly, former Supreme Court judge Arun Mishra had sparked controversy by publicly describing Prime Minister Modi as a "versatile genius" and an "internationally acclaimed visionary who can think global and act locally".

On a separate occasion, Prime Minister Modi visited the Supreme Court premises at the invitation of then-CJI Ranjan Gogoi on the sidelines of a BIMSTEC conference. Justice Gogoi was nominated to the Rajya Sabha following his retirement. Incidentally, Justice Gogoi was one of the four senior Supreme Court judges who held the unprecedented January 2018 press conference, spotlighting concerns over executive influence in case allocations under then Chief Justice Dipak Misra.

History shows that political friction with the judiciary carries 'unsettling' consequences. Following the landmark *Kesavananda Bharati* judgment, which established the basic structure doctrine limiting Parliament's amending powers, Indira Gandhi's government superseded Justices J.M. Shelat, K.S. Hegde and A.N. Grover to appoint Justice A.N. Ray as Chief Justice of India. The move triggered the immediate resignation of the three superseded judges.

Decades later, the imperative to insulate judicial appointments from executive overreach had culminated in the Collegium system and the adoption of the Bangalore Principles.

CISF to overhaul its administrative structure

The Hindu Bureau
NEW DELHI

The Central Industrial Security Force (CISF) is all set to overhaul its administrative and operational structure to strengthen specialised security management across critical infrastructure and strategic establishments under its protection.

The revised arrangements, approved by the Ministry of Home Affairs (MHA), will come into effect on November 1, the force said in a statement.

The reorganisation aims to align the force's supervisory structure more closely with the specific security requirements of different sectors, including power, coal, steel, petroleum and natural gas, ports, government buildings, museums, monuments and mass rapid transit systems.

The objective is to improve operational oversight, promote sector-specific expertise, ensure greater uniformity in security practices and strengthen coordination with the Ministries, Departments and installation managements.

The CISF provides security to a wide range of establishments, including airports, seaports, atomic energy and space installations, power plants, power grids, steel plants, coal mines, petroleum and natural gas facilities, currency note presses, government buildings, iconic monuments, and mass transit systems.

Each sector has distinct operational processes, infrastructure, potential hazards, and security vulnerabilities.


The revamp aims to align the force's supervisory structure with specific security requirements of different sectors. FILE PHOTO

Supervisory structure

Under the earlier arrangement, administrative and operational supervision was largely organised along geographical lines. Over the past 25 years, the CISF has progressively adopted a functional approach in selected sectors, including aviation and strategic establishments under the Department of Atomic Energy and the Department of Space.

After the review of its supervisory structure, the National Capital Region Sector has been reorganised into the Government Building Museums & Monuments and Mass Rapid Transit System (GBMM & MRTS) Sector, New Delhi. The North Sector (New Delhi) and North Eastern Sector-II (Kolkata) have been restructured as the Power-I and Power-II Sectors, respectively.

The Eastern Sector (Ranchi) has been transitioned into the Coal Sector, while the Central Sector (Bhilai) is now the dedicated Steel Sector. The North Eastern Sector (Guwahati) has been transformed into the Petroleum & Natural Gas Sector.

To strengthen security management at maritime trade infrastructure, the Western Sector (Mumbai) and the Southern Sector (Chennai) have been transformed into the West Coast Port Sector and the East Coast Port Sector, respectively.

The Department of Atomic Energy and the Department of Space Sector, Bengaluru, Airport Sector-I, New Delhi, and Airport Sector-II, Bengaluru, will continue under their respective functional arrangements.

The reorganisation assumes significance as the CISF's mandate expands beyond conventional industrial security to encompass new and specialised security responsibilities. The force has been designated as a Recognised Security Organisation (RSO) for port facilities under the International Ship and Port Facility Security (ISPS) Code. The functional realignment of port-sector supervision is expected to support the development of specialised port-security expertise and coordination with port authorities and other stakeholders.

The CISF has also been entrusted with security responsibilities at high-security correctional facilities in Jammu and Kashmir and designated as the nodal agency for aerial security of critical infrastructure. These responsibilities require specialised training, clear operational accountability and a coordinated approach to emerging security challenges.

SC bars mid-term fee hike in professional courses

Krishnadas Rajagopal
NEW DELHI

Much to the relief of families sacrificing their life savings for professional education, the Supreme Court has ruled that students enrolling in professional courses must know their exact expenses from the outset.

"A student entering a four-year or five-year professional programme like medicine or engineering should ordinarily know, at the very threshold, the financial liability that they or their family is undertaking," a Bench of Justices P.S. Narasimha and Alok Aradhe held.

The court said midstream fee hikes were impermissible except under unforeseeable or extraordinary economic conditions or statutory changes. Fee-fixation committees must function as continuing institutions, capable of anticipating costs, undertaking periodic assessments, and making timely decisions.

'Justified, not arbitrary'

Both the All India Council for Technical Education Regulations, 1994, and the National Education Policy, 2020, offer regulatory frameworks that pre-empt arbitrary fee increases by self-financing higher educational institutions while preserving legitimate fiscal space for them, the court added.

The verdict said the legitimate interest of students and parents to have certainty about the reasonable fee payable for the course must be balanced with the interest of self-financing educational institutions to secure adequate resources for providing quality education and improving standards. The court clarified that it was not against periodic fee revision.



"If such cost is not recovered from the students at the relevant time and is deferred to future batches, an inevitable revenue deficit would certainly arise and will either have a cascading effect on future batches or adversely affect the economic viability of the institution. It will have its own fallout on the infrastructure, quality of lecturers and professors, and also adversely affect student welfare," the judgment explained.

Fee certainty through efficient regulation is the paramount principle. There should be a sense of predictability about the financial liability of higher education. The authorities must notify the fee payable for the entire course before a student takes admission, the court said.

Fee disparity

The judgment, authored by Justice Narasimha, noted that National Sample Survey Office data showed that professional course fees were a staggering 711% higher than those for general academic programmes.

"For the overwhelming majority of households, the fee payable for a professional course constitutes a substantial part of the family's income and savings; it is not a matter of incidental expenditure," the court observed.

The court spoke for millions of parents who struggle to pay for their children's professional courses. Many families also rely on student loans to pay for higher education.

"Adverse variations in financial commitment, and the insecurities that it may create, will impact the student and the family, and, in the long run, have an adverse effect on the quality and character of the professionals that are in the making," the top court cautioned.

Oil firms bar ethanol from sugar for blending in fuel

The move, for the 2026-27 supply year, comes amid concerns over domestic sugar availability and rising prices after poor rainfall in key sugar-producing States such as Maharashtra and Karnataka

Press Trust of India
NEW DELHI

State-run oil marketing companies have barred ethanol made from sugar cane juice, sugar syrup and B-heavy molasses from their latest procurement tender for the 2026-27 supply year for blending in petrol, as the government seeks to protect domestic sugar supplies while sustaining its nationwide petrol-blending programme.

The tender, issued on October 16 by Bharat Petroleum Corporation Ltd. (BPCL) on behalf of oil marketing companies (OMCs), seeks bids for around 11 billion litres of ethanol for the supply year beginning in November to meet the requirement of doping 20% ethanol in petrol. It cuts the incentive for maize-based ethanol, lowering the procurement price to ₹68.86 a litre from ₹71.86.

"The Government of India has issued directions that sugar diversion for ethanol be reduced to zero for the ethanol supply year 2026-27," the tender document said. "No offers have been sought for ethanol produced from sugar cane juice, sugar or sugar syrup and B-heavy molasses in this quantity bid tender," it added.


The government has issued directions that sugar diversion for ethanol be reduced to zero for year 2026-27. GETTY IMAGES

The restrictions mark a shift in feedstock sourcing for the ethanol-blended petrol programme, which has relied on both grain-based and sugar cane-based ethanol to reduce crude oil imports and cut vehicle emissions.

The tender allows bids for ethanol produced from maize, broken rice, surplus rice procured by the Food Corporation of India (FCI), and C-heavy molasses, according to the document. The reduction in the maize incentive to ₹2.79 per litre from ₹5.79 is expected to lower returns for producers supplying ethanol from the grain, even as the exclusion of sugar-based feedstocks increases the importance of alternative raw materials.

The move comes amid concerns over domestic sugar availability and rising prices after poor rainfall in key sugar-producing States, including Maharashtra and Karnataka, raised the risk of lower output. India, the world's largest sugar consumer, has increasingly used ethanol blending to reduce its dependence on imported crude oil. However, diverting sugar cane and sugar for fuel production can compete with supplies available for food consumption, particularly when output is constrained.

The latest tender stresses the government's effort to balance its energy-security objectives with the need to maintain adequate sugar supplies and contain domestic prices. India has adopted petrol containing up to 20% ethanol, known as E20, as the nationwide standard as part of efforts to reduce oil import dependence and emissions.

Ethanol is produced from agricultural feedstocks, including sugarcane and grains, and blended with petrol to displace a portion of fossil fuel consumption.

The shift away from sugar-based ethanol in the 2026-27 procurement cycle is likely to increase the importance of grain-based production and make feedstock availability, procurement prices and distillery capacity key factors in meeting blending requirements.

Handle perishables only in controlled conditions: FSSAI

Press Trust of India
NEW DELHI

Food business operators must ensure strict adherence to hygiene standards, it says

The Food Safety and Standards Authority of India (FSSAI) has asked food business operators (FBOs) to transport perishable items under controlled-temperature conditions across the supply chain to safeguard public health. The food regulator has also asked State food safety officers to maintain strict vigilance and take action if rules are not followed.

In a social media post on Sunday, the FSSAI said it has issued an advisory on "handling, storage, and transportation of perishable foods like *khoa, paneer*, meat, and fish under controlled temperature conditions throughout the supply chain".

The FBOs must ensure strict adherence to hygiene and temperature standards, it added. The FSSAI mentioned that it has specified hygienic and sanitary requirements for FBOs under Schedule 4 of the Food Safety and Standards (Licensing and Registration of Food Businesses) Regulations, 2011.

"It has been brought to the notice of FSSAI that perishable food commodities are at times not handled, stored or transported under appropriate temperature conditions," the advisory noted. Because of their perishable nature, these food items are highly susceptible to microbial growth and spoilage if not maintained under appropriate temperature conditions.

"Protect the product from dust, moisture, pests, physical damage and cross-contamination during storage, loading, unloading and transportation," it told the FBOs.

Further, the FBOs have been told to maintain appropriate records and traceability of consignments. The regulator has also advised the commissioners of food safety at the state level, Indian Railways and Regional Directors of FSSAI to direct the licensing authorities and food safety officers concerned to maintain strict vigilance during inspections and transportation checks to ensure compliance with the law.

Study flags impact of Internet use on children

A. M. Jigeesh
NEW DELHI

A study on Internet use among school-going children, supported by the Indian Council of Social Science Research (ICSSR), recommends that digital literacy must be encouraged for children and parents with active support from schools.

Anxiety, sleep disturbances, and mood changes among children are the most reported health consequences, indicating that Internet use impacts children emotionally and physically.

Based on the findings, the study has recommended a "collaborative multi-stakeholder framework" to digitally empower children in the era of emerging technologies.

A report based on the study titled "Empowering Children in the Digital Age: Analysing Experiences and Vulnerabilities within the Framework of Vision Viksit Bharat and Sustainable Development Goals" has found 97.6% children accessing the Internet from home, with smartphones as the primary device.


Anxiety, sleep disturbances, and mood changes among children are the most reported health consequences. REUTERS

"Most of the children are spending 4.5 hours per day using the Internet. Their primary purpose for using the Internet is studying, followed by playing online games, watching videos, and talking to friends," Moina Khan, faculty with Bennet University, Uttar Pradesh, which conducted the study for ICSSR, said.

For the study conducted in 2025, Ms. Khan interviewed over 2,000 children in the 9-14 years age group studying in schools in Delhi, and about 500 parents.

"Children, schools as well as parents reported various benefits of digital platforms, including social media for children, from information, upskilling and community-building to collaborative learning and creative expressions. Children confidently use digital spaces exhibiting high technical or functional skills. However, what they lack are critical skills," Ms. Khan said.

Living in a bubble

She pointed out that at home, adults think children are tech savvy and better than them. "However, we are living in a bubble thinking they know it all. What they do not understand is systematic power manipulation and persuasion, algorithmic influence, and exploitation. The majority of children, irrespective of their perceived digital skills, reported feeling lonely when they did not use the Internet for long periods, experiencing anger when they did not find what they wanted online, and struggled to stop using the Internet even when they wanted to, reflecting signs of addiction," she added.

Nearly 47% children mentioned talking to unknown people in the past six months while many played games online with strangers, according to the study. "Talking to strangers online, playing games with them, and watching content inappropriate for their age was more common in the age group of 12-13 years.

A few reported experiencing cyberbullying and impersonation. Most of the children have mentioned not reporting uncomfortable experiences to parents or teachers. However, many of them considered parents as a preferred choice for reporting such incidents," the report added.

The report also indicates gender-based differences in Internet use, with boys generally using the Internet more than girls.

"Boys' greater social freedom often leads to less controlled Internet use, increasing their exposure to harmful content and online risks. Girls' restricted access could create a different threat by limiting their digital literacy and self-protection skills," it said.

What if a cheque bounces?

Sincy Wilson
Vaishali R. Venkat

In this digital era, cash is gradually losing its sheen while digital payments are rising. Yet, in certain high-value transactions such as property purchases, business deals and loan repayments, cheque continues to hold its ground. A cheque is a bridge between a promise and a payment. When the bridge collapses, the promise is breached, money fails to reach its destination, giving rise to financial and legal consequences.

Let's address some rudimentary aspects. To mitigate risk of disputes, both parties must verify the cheque's particulars and check for errors or unauthorised alterations. The issuer must ensure sufficient balance when the cheque is presented and avoid issuing signed blank cheques.

However, a cheque may still be dishonoured by the bank, that is, payment may be refused and the cheque returned unpaid, a situation known as a cheque bounce. Generally, when a cheque bounces, the bank provides a return memo to the recipient stating the reason. The recipient must retain the memo and notify the issuer to resolve the payment issue. A cheque may bounce for several reasons, including insufficient funds, account closure, stop-payment instructions or signature mismatches. The legal consequences rest on reason for dishonour and circumstances.

Legal consequences

A cheque bounce may be treated as a criminal offence under Section 138 of the Negotiable Instruments Act, 1881, punishable with imprisonment of up to two years, a fine of up to twice the cheque amount or both, if the statutory conditions are met.

Under Sections 118 and 139 of the Act, the court presumes, unless the contrary is proved, the cheque was issued for consideration and towards a legally enforceable debt or liability. The drawer (or accused) can rebut this by raising a probable defence that challenges the existence of the legally enforceable debt or liability.

For instance, Kamal issues Priya a cheque for ₹50,000 which is dishonoured due to insufficient funds. The court initially presumes Kamal owed Priya ₹50,000 and issued the cheque only to repay that debt. Kamal can challenge this by presenting evidence that casts doubt on the existence of his liability to Priya. However, he does not have to prove his case entirely on his own; the court considers proofs from both sides before arriving at a conclusion.

Legal timeline

The payee can initiate proceedings under Section 138 only if the cheque was presented within validity period, usually 3 months from its date. If the cheque bounced, the payee, within 30 days of receiving information from the bank, must send a written demand notice to the issuer. The defaulter has an option to settle the payment within 15 days from the date of receipt of the payee's demand notice. If payment is not made within the stipulated time, the payee might file a written complaint before the competent Magistrate within a month (30 days) after the 15-day notice period expires.

The court may accept a late complaint if the complainant provides sufficient reason for the delay. The parties may also settle the case by mutual agreement as Section 147 of the Negotiable Instruments Act, 1881, permits such cases to be settled amicably. Further, the recipient may approach a civil court to recover the unpaid amount. For eligible claims, a summary suit offers a procedure to speed up the recovery of certain debts. Under the Limitation Act, 1963, a money-recovery claim usually must be filed within three years though the date from which this period begins depends on the nature of the claim.

(Dr. Sincy Wilson is an Assistant Professor of Law, Vels University (VISTAS). Dr. Vaishali R. Venkat is an NISM & CRISIL-certified Wealth Manager and certified in NISM's Research Analyst module)

Two-face molecule could help make IBD treatments more effective

A drug that targets an overactive immune response may also weaken the gut's ability to fight harmful bacteria; understanding where the molecule is helping or hurting could be important before this pathway is considered for treatment, new research from National Institute of Immunology has found

Anirban Mukhopadhyay
NEW DELHI



As the protein RelB acts as both guardian and saboteur depending on its neighbourhood, safely targeting it remains a complex challenge. Representative image. SORA SHIMAZAKI/UNSPLASH

THEGIST

To execute many of its immune effects, the gut relies on IL-23, a signalling molecule that triggers inflammation to ward off threats; IL-23 functions as an alarm system at barrier surfaces such as the gut, skin, and lungs, a professor of chemistry and biochemistry said

RelB does not act as a heroic shield everywhere in the gut; while it builds a protective army in the intestinal wall, it suppresses regulatory T-cells — the immune system's peacekeepers — in the nearby lymph nodes

Researchers found that RelB activates production of RBP-J, a protein that unlocks a separate communication network; this connection enables immature cells to develop into the gut's IL-23-producing sentinels, revealing that RelB and the communication network cooperate to arm the gut

Deep within the intestinal wall, beyond where digestion and nutrient absorption occur, are immune outposts called solitary isolated lymphoid tissues, or SILTs. These structures host cells that govern how aggressively the gut reacts to microbes. A weak response lets pathogens take root while an overactive one causes the immune system to attack the gut – a hallmark of inflammatory disorders like Crohn's disease.

A new study in *Proceedings of the National Academy of Sciences* has traced this balancing act to a protein called RelB. By engineering mice lacking RelB in these cells, researchers at the National Institute of Immunology (NII), New Delhi, uncovered a paradox: the same protein drives opposite effects depending on its cellular environment and anatomical niche.

The finding could matter for patients with inflammatory bowel disease (IBD) because a drug that targets an overactive immune response may also weaken the gut's ability to fight harmful bacteria. And understanding where RelB is helping or hurting could be important before this pathway is considered for treatment.

The gate-keeper

To execute many of its immune effects, the gut relies on IL-23, a signalling molecule that triggers inflammation to ward off threats. IL-23 "functions as an alarm system at barrier surfaces such as the gut, skin, and lungs," Gourishankar Ghosh, a professor of chemistry and biochemistry at University of California, San Diego, U.S., said.

In the gut, it sparks a chain reaction that switches on antimicrobial defences. "However, uncontrolled production can drive chronic inflammatory diseases, including psoriasis and inflammatory bowel disease."

The researchers found RelB is a gatekeeper for this system: it dictates whether the gut can even produce the sentinel dendritic cells residing in SILT outposts to make IL-23. In RelB-deficient mice, this defender population nearly vanished.

Soumen Basak, an immunologist at NII who led the study, explained that earlier research suggested RelB simply helped existing cells pump out IL-23. Faced with the missing cells, the team investigated further.

"We considered two possible explanations: either the absence of... RelB reduced the number of IL-23-producing [dendritic cells] in these mice or simply a lack of RelB impaired the ability of these dendritic cells to make IL-23," Dr. Basak said. The study proved the former is the primary culprit.

Known troublemaker

However, RelB does not act as a heroic shield everywhere in the gut. While it builds a protective army in the intestinal wall, it suppresses regulatory T-cells – the immune system's peacekeepers – in the nearby lymph nodes.

Earlier work from Dr. Basak's lab found that by shutting down these peacekeepers, RelB worsens chronic intestinal inflammation.

The answer to how a single protein pulls off this Jekyll-and-Hyde routine might lie in the dendritic cells themselves.

Devesha Kulkarni, assistant professor of medicine at University of California, Los Angeles (UCLA), said dendritic cells are "not one fixed cell type" with a single job. Instead, they are shapeshifters that adopt distinct identities based on the tissue they lie in and the local chemical signals.

And now, scientists know RelB helps fix that local identity – a role Dr. Basak said he considers to be a major conceptual leap.

The researchers found RelB switches on the production of a second protein, RBP-J. This acts as a key, unlocking a completely different communication network. And only when this connection completes can immature cells develop into the gut's specialised IL-23-producing sentinels. The finding reveals a surprising twist: two systems long assumed to operate independently – RelB and the communication network – actually cooperate to arm the gut.

"Making a delicious dish requires the right combination of many ingredients – salt, turmeric, garlic, ginger, and spices," Dr. Basak said. "Each ingredient contributes a unique flavour, but none can produce the desired taste on its own. Our study emphasises that signalling pathways must also work together in a coordinated manner to generate appropriate cellular and immune responses."

That said, a deeper mystery remains: how can a single protein act as a brake in one context and an accelerator in another?

Alexander Hoffmann, professor of immunology at UCLA, said the contradiction comes down to competition. RelB, he said, is a relatively weak activator compared to its cousin, RelA, so when both proteins try to access the same gene, the weaker RelB gets in the way.

By blocking RelA's strong signal, RelB dials down the overall effect so "it seems like a repressor," Dr. Hoffmann explained.

But on genes that RelA cannot access – like the one that triggers the gut's defenses – "RelB's weak activation potential is sufficient to drive gene expression."

"This new work is a remarkable example of how the Basak lab over many years has developed expertise in dissecting complex networks of interaction," he added.

Delicate drug target

Drugs that block IL-23 are already frontline treatments for inflammatory bowel disease, making RelB an intriguing but delicate new drug target. Because the protein acts as both guardian and saboteur depending on its neighbourhood, safely targeting it remains a complex challenge.

Saurabh Mehandru, associate professor of Gastroenterology at the Icahn School of Medicine at Mount Sinai, New York, said that because the research used mice, "the findings will need to be extended before they can be read directly into patient care."

He also pointed to a key gap that Basak's team has also acknowledged: because the protective cell clusters are almost entirely missing in the modified mice, it is difficult to prove whether RelB is required to build those cells in the first place or simply to keep them alive once they are formed. Future experiments tracking the lifespans of these cells will ultimately settle that debate, Dr. Mehandru added.

Dr. Basak is already looking ahead. His team wants to know if the relay operates at other vulnerable barriers, like the lungs, and whether tweaking it could yield new treatments.

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BIG SHOT



Baby sea turtles crawl towards the sea on Iztuzu Beach on Turkey's Mediterranean coast. The loggerhead turtle, which nests on shores across the Mediterranean, is the mascot of COP31, the annual UN climate summit that Turkey will host next month. REUTERS

WHAT IS IT?

Satellite radiation: blocking views

Vasudevan Mukunth

Communications satellites in low-earth orbit are emitting radio signals that interfere with astronomy across a range of frequencies, a study published in *Monthly Notices of the Royal Astronomical Society* has reported.

The findings, reported under a satellite radiation monitoring programme called SNIFFLES, suggest protecting the interests of astronomers requires operators to regulate licensed transmissions as well as stray radiation from onboard electronics.

The study authors used the Mopra radio telescope and the Australia Telescope Compact Array to track satellites of the Starlink, OneWeb, and China's Guowang constellations across nearly 376 hours from April 2024.

Based on their observations, they classified the radiation as intended, unwanted, and unintended. Intended transmissions included direct-to-cell signals from Starlink satellites. Unwanted transmissions included emissions from transmitters, including signals at multiples of the transmission frequency (harmonics). Unintended radiation included stray emissions from electronic components, such as power converters and computers.

The authors also found a signal from the OneWeb constellation at 1,613.19 megahertz (MHz) that could get in the way of telescopes studying hydroxyl ions ($-OH^-$) in space. They also reported radiation at 2,700 Hz appearing in nearly 77% of observations from a specific version of Starlink satellites.



A view of radio telescopes of the Karl G. Jansky Very Large Array in the U.S. YIFU WU/UNSPLASH

Some signals were also 100-billion-times brighter than typical astronomical sources, which the authors warned could overwhelm telescopes.

The study's authors concluded that operators could adopt existing international rules about radio transmissions through the atmosphere to control emissions from satellites in earth orbit. They also called for operators, satellite designers, and space agencies to continuously monitor emissions, especially as direct-to-cell services expand via private enterprises like Starlink and Amazon Leo.

For feedback and suggestions
for 'Science', please write to **science@thehindu.co.in** with the subject 'Daily page'